University of Mumbai

UG Project on Topic

Comparative Study of FEA and Theoretical Calculations of Design of Pressure Vessels Using ASME Codes

Submitted in partial fulfilment of the requirements

For the degree of

Bachelor of Technology

SUBMITTED BY – GROUP NO. 30

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Certificate

This is to certify that the Topic Seminar Report on "**COMPARATIVE STUDY OF FEA AND THEORETICAL CALCULATIONS FOR THE DESIGN OF PRESSURE VESSELS USING ASME CODES**" is bonafide record of the seminar work done by Akshay Mohan, Balaji Venkatesan, Keyur Joshi and Rahil Doshi in the year 2019-2020 under the guidance of Prof Priyanka Patil, Department of Mechanical Engineering in partial fulfilment of the requirements for the Bachelor of Technology degree in Mechanical Engineering of the University of Mumbai.

____________________ Guide / Co-Guide

____________________ Head of the Department

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Date:
Place: Mumbai-77

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Certificate of Approval of Examiners

We certify that this Special Topic Report and Seminar on **"COMPARATIVE STUDY OF FEA AND THEORETICAL CALCULATIONS FOR The DESIGN OF PRESSURE VESSELS USING ASME CODES"** is a bonafide record of seminar work done by Akshay Mohan, Keyur Joshi, Balaji Venkatesan and Rahil Doshi for partial fulfilment of the requirements for the Bachelor of Technology degree in Mechanical Engineering of the University of Mumbai.

External Examiner / Expert	Guide / Internal Examiner

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DECLARATION

We declare that this written report submission represents the work done based on our and / or others' ideas with adequately cited and referenced original source. We also declare that we have adhered to all principles of academic honesty and integrity as we have not misinterpreted or fabricated or falsified any idea/data/fact/source/original work/ matter in our submission.

We understand that any violation of the above will be cause for disciplinary action by the college and may evoke the penal action from the sources which have not been properly cited or from whom proper permission is not sought.

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Abstract

Pressure vessels are one of the most important components used in many industries ranging from process to oil and gas. Traditional design of pressure vessel involved design by calculations and relied on the designer's experience. Modern design is done in accordance with regulatory guidelines, described in many international codes. Most commonly used codes for design of pressure vessels are ASME BPVC codes. This project discusses the design of vertical pressure vessels and the calculations for the same using ASME guidelines. This project was executed with the help of Geecy engineering private limited which specializes in the design and fabrication of boilers and pressure vessels.

After the design parameters were obtained using ASME codes, the solid model of the same was developed using CAD software. Further, the Finite element analysis of the developed model was done using commercially available FEA packages. The results are to be evaluated and the comparison of the same is to be done with those obtained from hand calculations. Further It can be evaluated using FEA that whether the vessel maintains structural integrity at thicknesses below those calculated by traditional methods, thereby optimizing the design

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Chapter 1

Introduction

This chapter presents the brief introduction of pressure vessels, necessity of ASME codes in design, types of pressure vessels and their components and the objectives and methodology of the project.

1.1 Background

Pressure Vessels are structures or containers designed to hold gases, vapours or liquids at high pressures. Vessels, tanks or pipelines that carry store or receive fluids are called pressure vessels. It is a container with pressure differential between the inside and the outside. Generally, pressure vessels are designed such that they can sustain a high internal pressure. Pressure vessels are an important component in all process industries and find widespread application in chemical, power, pharmaceutical and oil and gas industries.

Modern pressure vessels were first built at the onset of the industrial revolution. It was during this period that humanity realized the driving power of steam and hence various equipment were conceived which were necessary for production, storage and transport of steam at high pressures, like boilers and pressure vessels.

Pressure vessels were traditionally designed using systematic design by formula approach which was based largely upon the designer’s experience and simple mechanics. In the Design by formula, the vessel geometry and major dimensions such as radius, length, etc. were specified and the required thickness was then calculated for a given load using mechanical and empirical equations and graphical data.

In the early part of the twentieth century, numerous industrial catastrophes involving pressure vessels occurred which caused a loss of life and property. This gave rise to a standardized set of regulatory norms for the design of pressure vessels, like those developed by American Society of Mechanical Engineers (ASME) These codes involve better scientific practices like design by analysis; in which a designer analyses various stresses and compares them with code standards to ascertain the safety.

New technologies like Computer aided engineering, Simulation, finite element Analysis, non-destructive testing methods etc. have given new methods to design and analyse pressure vessels and to develop safer equipment for the industry. Our thesis also discusses the use of FEA techniques in concurrence with analytical calculations using ASME codes for analysis of pressure vessels.

1.2 Objectives

- Study of the basics of pressure vessel and literature review
- Study of ASME Code for Pressure Vessels (ASME BPVC SEC VIII).
- Hand Calculations for the thickness of the shell, thickness of dished end, nozzles and support.
- Excel Sheet development for pressure vessel (ASME SEC.VIII DIV 1).
- Modelling of the components for Division 1 (low-pressure vessel).
- FEA analysis of Static Burst pressure.
- Validation of results with hand calculations.

1.3 Motivation/ Need for ASME codes

Pressure vessels are containers made to store pressurized media and as such, have inherent safety risks. The safe application of these equipment has always been an important challenge. A large loss of life and property resulted due to many catastrophic accidents that occurred in the early twentieth century. This led the policymakers and engineers to develop a standard code for the design of pressure vessels that would ensure the structural integrity and provide engineers a framework for safe design.

It was American Society of Mechanical Engineers (ASME) that stepped in to unify the design formulation and make standard specifications for steam boilers and pressure vessels.

After more years of research and development ASME published the first edition of the BPVC Code in 1914. The code was called ASME Rules of Construction of Stationary Boilers and for Allowable Working Pressures. The code developed over time into the ASME Boiler and Pressure

Vessel Code (ASME BPVC), which today has over 92,000 copies in use, in over 100 countries around the world. Nowadays, the code consisted of 16,000 pages in 28 volumes and is presented in eleven sections, with multiple subdivisions, parts, subsections, and Mandatory and non-mandatory appendices

Section VIII is the most used code for design and construction of almost all pressure vessels in the United States and also all over the world.

In the following thesis, the calculations are done for the design of pressure vessels based on section VIII, Division 1 of the BPVC. This division covers the mandatory requirements, specific prohibitions and non-mandatory guidance for materials, design, fabrication, inspection and testing, markings and reports, overpressure protection and certification of pressure vessels having an internal or external pressure which exceeds 15 psi (100 kPa).

1.4 Pressure vessel types and components

Pressure vessels can be classified according to dimensions as:

If the ratio of diameter to thickness is less than 10, it is called as the shell pressure vessel, and if the ratio D/t is more than 10, it is called thick shell pressure vessel.

Examples of thin cylinders include pipes, storage tanks while pressure cylinders and gun barrels are an example of thick shell cylinders.

Pressure vessels can be classified according to shape as:

Pressure vessels can theoretically be made in any shape but the most commonly used are cylindrical and spherical pressure vessels. A common design would be a cylinder with end caps called heads. Head shapes are commonly ellipsoidal or dished (tori spherical) more complicated shapes can be fabricated but are usually not used much as they are difficult to analyse and construct.

Materials with good tensile strength and chemical stability in the given application are selected to manufacture pressure vessels. In addition to adequate mechanical strength, current standards dictate the use of steel with a high impact resistance, especially for vessels used in low temperatures. In applications where carbon steel would suffer corrosion, special corrosion-resistant material should also be used. The range of materials used for pressure vessels is wide and includes, but is not limited to, the following:

1. Carbon steel (with less than 0.25% carbon).
2. Carbon manganese steel (giving higher strength than carbon steel).
3. Low alloy steels.
4. High alloy steels.
5. Austenitic stainless steels.
6. Non-ferrous materials (aluminium, copper, nickel and alloys).

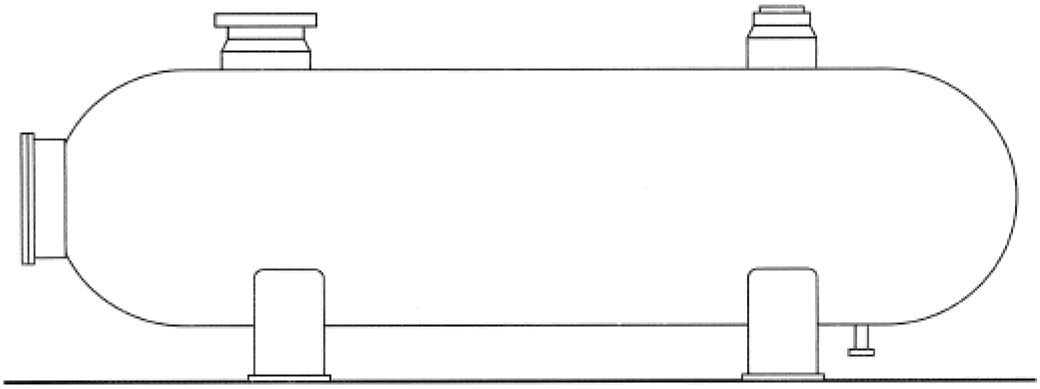

Fig. 1 Schematic of a pressure vessel showing various parts

Terminologies in Pressure vessel design

Shell: The Shell contains the pressure and consists of plates that have been welded together with an axis. Horizontal drums use shells with a cylindrical shape.

Dished end: This is what closes off the end of a pressure vessel. Curved heads have less weight, cost less and have more strength than flat heads.

Hoop stress: This is the stress which is set up in resisting the bursting effect of the applied internal pressure. The hoop stress is the force exerted circumferentially (perpendicular both to the axis and to the radius of the object) in both directions on every particle in the cylinder wall.

Longitudinal stress: Longitudinal stress is the stress in a pipe wall, acting along the longitudinal axis of the pipe. And it is produced by the pressure of the fluid in the pipe.

Operating pressure: The Operating pressure is the gauge pressure which prevails inside equipment and piping during any intended operation. The Operating pressure is determined by the process engineer.

Operating temperature: The operating temperature is the temperature that prevails inside equipment and piping during the predominant intended operation

Nozzles: The tie-in connection between the vessel or equipment and the piping system. Nozzles are provided in locations where a commodity is either introduced or removed from a vessel or piece of equipment.

Manholes: Similar to large nozzles that allow workers entry points into a vessel. They generally are 18 ID and are accessible by ladders and platforms. When not in use, the manhole is sealed with a blind flange.

Skirt support: Vessels with height and vertical pressure use skirts. These are welded to the shell section with enough length to go to the bottom of the head. This adds flexibility and prevents high stress thermally.

Saddle support: These give support to horizontal drums in two locations and reduce stress.

1.5 Methodology

The basic methodology of the project involved working with Geecy engineering private limited for theoretical calculations and the FEA analysis was performed on the college campus using software licenses provided by the university. The first step in the process was interaction with the guide in the industry to understand and define basic pressure vessel parts and terminology. After that, client given data was studied to determine the requirements for the design. These are included under functional and operational requirements. Further, study of ASME codes under Sec VIII Division 1 was undertaken. The various parameters provided by the client included data such as vessel length, diameter, operating pressure and temperature. Once the design conditions were established, the material selection was done. The recommended material by the industry guide was SAE 240 and alloy S31803.

Upon selection of material, its various structural properties were taken in to account by referring to ASME materials appendix. The material properties were used in further calculations. The preliminary calculations involving the codes UG 27, UG 28 and UG 32 were carried out, in which the motive was to calculate the thickness of shell and dished end. The subsequent calculations were for determining the reinforcements for openings in the vessel and stresses in various sections of the skirt support. The ASME guidelines involve stress analysis in various parts, which are discussed in detail in the chapter on theoretical calculation. After the stress analysis and design check, the pressure vessel would be fabricated, inspected and delivered to the client.

In the latter part of the project, the pressure vessel was modelled using Solidworks and the static structural analysis was carried out in Ansys 2019. The analysis was carried out in different steps. The first step was to include only the shell and dished end and to analyse for burst pressure. The subsequent iterations included skirt support and openings in the nozzle to further replicate the model as close to real vessel as possible, but with certain assumptions, to get accurate results.

Finally, the results of various iterations were compared with those obtained from theoretical calculations.

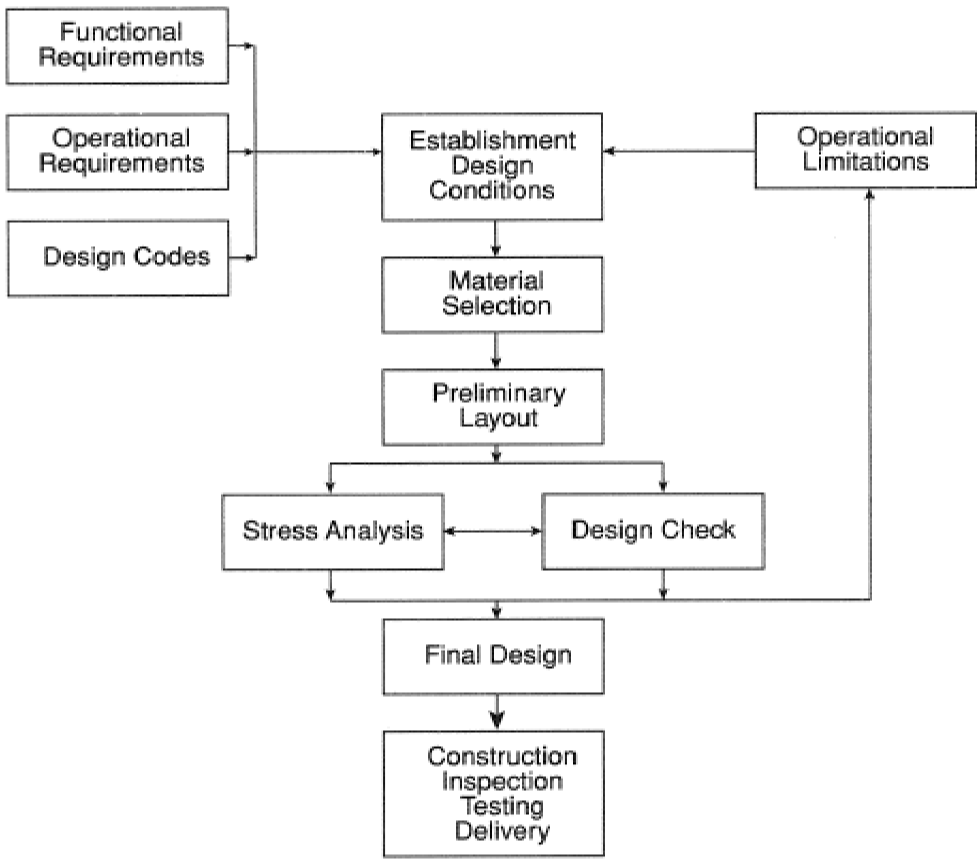

Fig. 2 Methodology flow chart

Operational Requirements:

The first step in this design procedure is to set down the operational requirements. These are imposed on the vessel as part of the overall plant and include the following:

- **Operating pressure**: As well as the normal steady operating pressure, the maximum maintained pressure needs to be defined. Regulations and/or standards will define how this maximum pressure is translated into vessel design pressure.
- **Fluid conditions**: Maximum and minimum fluid temperatures will need to be specified and translated into metal design temperatures. Fluid physical and chemical properties will influence material choice and specific gravity will affect support design.
- **External loads**: Loads to be considered include wind, snow, and local loads such as piping reactions and deadweight of equipment supported by the vessel.

- **Transient conditions**: Some vessels may require an assessment of cyclic loads resulting from operational pressure, temperature, structural and acoustic vibration loading.

Functional Requirements

Next the functional requirements, which cover geometrical parameters, are defined. Some of these parameters are again defined by the plant design team whilst some are left to the discretion of the pressure vessel designer. The functional requirements include the following:

- Size and shape of the vessel.
- Method of vessel support.
- Location and size of attachments and nozzles.

Chapter 2
Literature Survey

This chapter presents the literature survey about the design of pressure vessels and use of finite element analysis for the same. It also includes the research gap based on previous research on the topic

2.1 Literature Gap

Table 1

Table 1 Research Papers and the knowledge gaps

Sr. No	Name of Author / Year of Publication	Description	Research paper gap
[1]	Puneet Deolia, Firoz A Shaikh	1) Finite element analysis to estimate burst pressure of mild steel pressure vessel. 2) This paper deals with the use of material curves derived by Ramberg Osgood equation for predicting bursting value of pressure vessels. 3) The model accurately captures material curves in strain hardening region, and predicts burst pressure using FEA methods. 4) FEA results are compared with experimental outcomes. 5) It is found that Ramberg Osgood material model can be used for any material which has information regarding yielding limit and rupture. 6) This will eventually save the time and cost of actual testing.	1)Due to variation in the material properties of material used in the experimentation, Ramberg Osgood model shows some discrepancy in prediction of actual burst pressure values.

Sr. No	Name of Author / Year of Publication	Description	Research paper gap
[2]	Hsoung-Wei Chou, Chin-Cheng Huang 2017	1. Study of ASME section X1 for design of PWR (pressurised water reactors) pressure vessel in nuclear power plant in Taiwan. 2. Evaluation of new pressure-temperature curves apart from those in ASME section XI. 3. Risk informed methods of P-T transient methods were developed. 4. Simulations using Probabilistic Fracture Mechanics (PFM) were done based on Monte Carlo FAV PFM software. 5. Probability of crack initiation and penetration were estimated for various reactor pressure vessel (RPV) materials. 6. Based on PFM results, domestic PWR was found to have structural integrity if P-t curves were updated to those found by risk informed method.	

Sr. No	Name of Author / Year of Publication	Description	Research paper gap
[3]	B.S. Thakkar, S.A. Thakkar 2012	Objectives were: 1. Scope of ASME codes were studied 2. Various international codes apart from ASME were presented 3. Modes of failure like fatigue, ductile tension was discussed 4. Safety precautions in installation and maintenance of pressure vessels were discussed 5. Various sections in ASME BPVC section VIII and their subsections like UG 32, UG 45 etc were explained. 6. CAD Modelling of shell and hemispherical end was done on Pro E software.	1. Further analysis of the model developed can be done with commercial FEA packages. 2. Design of supports lake saddle and skirt support can be included Calculations for nozzle and nozzle pad can be done.

Sr. No	Name of Author / Year of Publication	Description	Research paper gap
[4]	Arun Kumar, H.S. Manjunath, Amith Kumar. Nov -2017.	1. The main objectives of this research paper were: • Thickness calculation by ASME codes • Geometrical built up of the problem • Finite element optimization 2. Two pressure vessel systems are analysed using Finite element analysis and compared with ASME stress conditions. 3. Design optimization is carried out using ANSYS software. 4. Design of pressure vessel major dimensions for the given pressure load and design optimization of the weld region is the main definition of the problem.	1. No voids and cracks are assumed in the problem. 2. Analysis is carried out with in the yield point of the material. 3. Connections are assumed to be complete and load transfer is proper. **Further research opportunities: -** 1. Analysis can be continued with change of material for optimization 2. Spectrum loading can be considered for the problem. 3. Effect of defects in the welds can be considered to find the strength. 4. Possible thermal effects can be considered. 5. Fluid turbulence effect inside the pressure vessel can be considered.

[5]	P. K. Nziu and L. M. Masu International Journal of Mechanical and Materials Engineering (2019)	1. The aim was to establish the effect of cross bore configuration geometry on stress concentration in cross-bored high-pressure vessels. 2. Due to the discontinuities in the cylinder, the stress distribution along the cylinder wall is not uniform. These discontinuities create regions of high stress that are referred to as stress concentrations. 3. The openings introduce geometric discontinuities that alter the uniform stress distribution in the cylinder walls. The geometric discontinuities create regions of high stress concentration, especially near the openings. 4. In cross-bored cylinders, the SCF depends on the geometric configuration of the cross bore. The parameters of cross bore geometry with a major effect on stress concentration include cross bore size, shape, angle of obliquity, and thickness ratio. 5. Cross bore size The results showed an increase in SCFs as the ratio of cross bore to main cylinder bore increases. For a particular thickness ratio, the SCF increases with increasing cross bore size. 6. Cross bore shape Incorporating chamfers, blend, or radius entry on circular cross bore, cause stress redistribution that leads to a reduction in SCF. A SCF reduction of up to 34.2% was noted at the main bore due to the introduction of chamfers. 7. Cross bore obliquity The study reported increased mesh element distortion whenever the obliquity angle was below 30°. It	1. Despite close interrelation of the geometric parameters, the majority of the reviewed studies addressed each parameter separately. 2. studies on optimization of cross bore geometry parameters have not been adequately investigated. 3. the accuracy of numerical solutions depends on the correct usage of the type of element, mesh density, accurate modelling of the material, loading, and boundary conditions. 4. In analytical methods, stress analysis is done using elastic, elastoplastic, or plastic theories. However, the accuracy of the analytical method depends on the assumptions made during the development of the theory. 5. It is evident that there is no universally known and accepted method for determining optimum stress concentration factors in thick-walled pressure vessels, considering the effects of the combined various geometric design parameters identified. 6. the existing solutions addressed the optimum conditions based on each design parameter separately, despite most of the parameters being closely interlinked. Other

Sr. No	Name of Author / Year of Publication	Description	Research paper gap
		was observed that, as the oblique angle reduced from 90° to 30°, the SCF magnitude increased significantly. 8. Thickness ratio The study reported that SCFs decrease with decreasing thickness ratio. Thick-walled cylinders were more suitable for chamfering than thin-walled cylinders.	authors compared magnitudes of stress concentrations without taking into consideration the size of the cross bore. 7. studies carried out so far have failed to determine the optimal conditions in high-pressure vessels with a cross bore under the combination of static, thermal, and dynamic stresses, arising from the geometric configuration, working fluids, at high pressures and temperatures, despite this being a common phenomenon in practice.

Sr. No	Name of Author / Year of Publication	Description	Research paper gap
[6]	B. Siva kumara , P. Prasanna , J. Sushma , K.P. Srikanth ICMPC 2017	1. The main objective of this paper is to design and analysis of pressure vessel. The stress development in the pressure wall are analysed by using Ansys and an optimized model is modelled to overcome the stresses produced in the vessel. 2. When the pressure vessel is exposed to pressure, the material is subjected to pressure loading, and stresses, from all directions. The normal stresses resulting from this pressure are functions of the radius of the element, the shape of the pressure vessel and the applied pressure. 3. In developing the design, a number of criteria must be considered such as the properties of materials used, the induced stresses, elastic stability, and the aesthetic appearance of the unit. 4. The internationally accepted for design of pressure vessel code is American Society of Mechanical Engineering (ASME). 5. We observed that the stress and deformed values of the modified model is better than the actual model. Because of the reduction of the stress values the life time of the vessel may increase.	1. Numerical methods give approximate results that are fairly accurate. However, most of the commercial software’s are predesigned with limited provisions for any alterations.

Sr. No	Name of Author / Year of Publication	Description	Research paper gap
[7]	M. J Mungla, K. N Srinivasan, Prof. V R Iyer (2013)	1. The main objective of this paper is to design and analysis of various components of pressure vessels like shell, flanges, nozzle and support structures along using ASME codes. 2. Calculations are done separately, considering wind loads and seismic loads, one at a time.	1. Design of base-ring and skirt sections has not been covered under ASME code and their dimensions are calculated with general design principles.

To conclude, we have satisfied the objectives stated at the start of Literature Review. The information received from every paper and the knowledge gap identified has been stated alongside each of the studied papers. This shall help us in deciding the further course of our project.

Chapter 3

Theoretical Calculations using ASME Codes

This chapter presents the analytical calculations performed for the design of vertical pressure vessel. The calculations performed are under sections UG27, UG28, UG32, UG33, UG36, UG37 and UG40

ASME codes introduction:

The ASME Boiler & Pressure Vessel Code (BPVC) is an American Society of Mechanical Engineers (ASME) standard that regulates the design and construction of boilers and pressure vessels. ASME Section VIII is a section of the code that is dedicated to boiler and pressure vessels and hence called BPVC (boiler pressure vessel code) It contains details about design, fabrication, inspection, testing etc of pressure vessels. It refers specifically to those vessels which have their operating pressure above 15 psig. Section VIII is divided into 3 divisions, each of which cover different specifications. Division 1 is concerned with requirements for design, fabrication, testing, certification etc. Division 2 contains standards for non- destructive testing and materials. Division 3 includes guidelines for pressure vessels operating above 10000 psig.

ASME BPVC Section II - Materials

The section of the ASME BPVC consists of 4 parts.

Part A - Ferrous Material Specifications

This Part is a supplementary book referenced by other sections of the Code. It provides material specifications for ferrous materials which are suitable for use in the construction of pressure vessels.

The specifications contained in this Part specify the mechanical properties, heat treatment, heat and product chemical composition and analysis, test specimens, and methodologies of testing. The designation of the specifications starts with 'SA' and a number which is taken from the ASTM 'A' specifications.

Part B - Nonferrous Material Specifications

This Part is a supplementary book referenced by other sections of the Code. It provides material specifications for nonferrous materials which are suitable for use in the construction of pressure vessels.

The specifications contained in this Part specify the mechanical properties, heat treatment, heat and product chemical composition and analysis, test specimens, and methodologies of testing. The designation of the specifications starts with 'SB' and a number which is taken from the ASTM 'B' specifications.

Part C - Specifications for Welding Rods, Electrodes, and Filler Metals

This Part is a supplementary book referenced by other sections of the Code. It provides mechanical properties, heat treatment, heat and product chemical composition and analysis, test specimens, and methodologies of testing for welding rods, filler metals and electrodes used in the construction of pressure vessels.

The specifications contained in this Part are designated with 'SFA' and a number which is taken from the American Welding Society (AWS) specifications.

Part D - Properties (Customary/Metric)

This Part is a supplementary book referenced by other sections of the Code. It provides tables for the design stress values, tensile and yield stress values as well as tables for material properties (Modulus of Elasticity, Coefficient of heat transfer.)

ASME BPVC Section VIII - Rules for Construction of Pressure Vessels

The section of the ASME BPVC consists of 3 divisions.

ASME Section VIII Division I

This division covers the mandatory requirements, specific prohibitions and nonmandatory guidance for materials, design, fabrication, inspection and testing, markings and reports, overpressure protection and certification of pressure vessels having an internal or external pressure which exceeds 15 psi (100 kPa).

Division 2 - Alternative Rules

This division covers the mandatory requirements, specific prohibitions and nonmandatory guidance for materials, design, fabrication, inspection and testing, markings and reports, overpressure protection and certification of pressure vessels having an internal or external pressure which exceeds 3000 psi (20700 kPa) but less than 10,000 psi.

Division 3 - Alternative Rules for Construction of High-Pressure Vessel

This division covers the mandatory requirements, specific prohibitions and nonmandatory guidance for materials, design, fabrication, inspection and testing, markings and reports, overpressure protection and certification of pressure vessels having an internal or external pressure which exceeds 10,000 psi (70,000 kPa).

The following calculations were performed by us with the help of our guide at Geecy Engineering private limited. These involve various sections under ASME BPVC sec VIII.

ASME BPVC VIII.I

Assumptions taken into consideration while performing the hand calculations were:

1) Joint efficiency is considered to be 100%.
2) Corrosion Resistance is neglected.
3) The pressure vessel is considered thin.
4) Circumferential and Radial stresses are assumed to be constant.

UG 27: Thickness of shells under internal pressure

E = Joint efficiency

P = internal design pressure

R = inside shell radius

S = maximum allowable stress

t = minimum required thickness of shell

Cylindrical shells

1. Circumferential stress

Formula: t=PR/(SE-0.6P)

Applicable when:

a. t<=R/2
b. P<=0.385(SE)

Given Customer Data

1. Fluid circulated - low pressure steam
2. Design pressure - 90/14.5 psi
3. Maximum design temperature - 400/400 F
4. Corrosion allowance - 0/0 inch
5. Main shell material - SA-240 UNS S31803

From ASME BPVC II D MATERIALS,

For material SA-240 and alloy S31803, at 400 F,

S = 164.644 MPa = 23879 psi and E = 1

Hence, 0.385(SE) = 9193.415 psi

(P<=0.385(SE) condition satisfied)

t = [90*41.5/2]/23879-0.6*90 = 0.078384 inch

t = 1.99 mm

1. Longitudinal Stress

Formula: t=PR/(2*SE+0.4P)

Applicable when: P<=1.25(SE)

Hence, 1.25(SE) = 29848.75 psi

(P<=1.25(SE) condition satisfied)

t = [90*41.5/2]/2*23879+0.4*90 = 0.0739 inch

t = 0.99 mm

Hence, selecting standard thickness value as per industrial requirements,

t = 6 mm = 0.23622 inch

UG 28: Thickness of shell and tubes under external pressure

From ASME BPVC II D MATERIALS,

Subpart 3: factors A and B are determined

DATA: L = 75.2 inch

Do = Di + 2t = 41.5 + 2*0.2362 = 41.9724 inch

Hence, L/Do = 1.7916 and Do/t = 177.6987

From pg. 794, Figure G

A = factor determined from figure G, subpart 3 of section II, part D

B = maximum design metal temperature

Do = external diameter of cylinder shell

A = 3.2*0.0001 = 3.2 * 10^4

From page 797, (y >207 MPa)

B = 33

Pa = maximum allowable external working pressure

Pa = 4B/[3*(Do/t)]

Hence, Pa = 0.24761 MPa = 35.9128 psi

Hence, $P_a > P_{external}$

Hence, we can proceed with t = 6mm.

UG 32: Formed heads, and sections, and pressure on the concave side.

For ellipsoidal heads,

Formula: t=PD/(2*SE-0.2P)

Hence,

t = [90*41.5/2]/2*23879-0.2*90 = 0.15647 inch

t = 3.9744 mm

Considering approximation of 2:1 ellipsoidal head,

Knuckle radius = 0.17 D = 7.055 inch

Spherical radius = 0.9 D = 37.35 inch

UG 33: Formed heads, pressure on convex side

Do/2h = ratio of major to minor axis of ellipsoidal head equal to the outside diameter of head skirt divided by twice the height of head.\

h_o = D/4 = 10.375 inch
$D_o/2h_o$ = 2.022

From table UG 33.1, corresponding to $D_o/2h_o$ = 2.022,
K_o = 0.9

Ro = Ko * Do = 37.775 inch

A = 0.125/[Ro/t] = $7.81664 * 10^{-4}$

From page 797, B = 80

P_a = B/[Ro/t] = 72.55 psi
Since, P_a is greater than P_o, hence t = 6mm is considered.

UG 36: Openings in pressure vessels

For vessels over 60 inches (i.e:1500mm), h is one third of vessel diameter

h = 41.5/3 = 13.833 inch.

UG-37: Reinforcement required for opening in Shells and formed heads

A) Without reinforcing element
B) Total cross- Sectional area of reinforcement requirement in the plane
under consideration (includes consideration of nozzle area through shelf if $S_n/S_v < 1.0$)
$A = d\, t_r F + 2\, t_n{*}t_r{*}\, F\, (1- F_{r1})$
d = finish diameter of circular opening
tr = Required thickness of seamless shell based on the hoop stress.
F = Correction factor that compensates for variation in internal pressure stresses on different planes
t_n = nozzle wall thickness
f_{r1} = Sn/Sv for nozzle wall inserted through the vessel wall
= 1.0 for nozzle wall abutting the vessel wall and nozzles shown in Figure
S_n = Allowable stress at design temperature

S_v = Material stress at a particular temperature

UG-40

sketch (j), (k), (n) & (o)

$A = dt_rF + 2t_nt_rF(1-f_{r1})$ [Area required]

For nozzle No3,

d = 16" = 16*25-4 = 406.4 mm [NPS16]

T_r = 1.99 = 2 mm

F =1.0

t_n = 10 mm

f_{r1}= Sn/Sv (since the material for nozzle and Cylindrical shell plate is the same)

E_1 = 1 when an opening is in the solid plate

= 0.85 when an opening is located in an autogenously welded pipe

t = 6 mm specified vessel wall thickness

$F_{r2} = S_n/S_v = 1.0$

f_{r3} = (lesser of S_n or S_p)/S_v = 1

$f_{r4} = S_p/S_v = 1$

A = 406.4*2*1 + 0 = 812.8 mm^2

A_1 = d(E1t-Ftr) - 2tn(E1t-Ftr) (1-fr1)

= 2(t+tn) (E1t-Ftr) - 2tn(E1t-Ftr) (1-fr1)

A_1 = 1625.6 mm^2

= 128 mm^2

Taking larger value of A1 = 1625.6 mm^2

$A_2 = 5(t_n-t_{rn})\ fr_2t$

$= 5(t_n-t_{rn})\ fr_2t_n$

For t_{rn} = PR/(SE-0.6P)

R = ½ (Finish Diameter of circular opening)

=½ (406.4) = 203.2mm

R = 8’’

t_{rn} = 90*8/ (23879*1-0.6*90)

$= 0.0302" = 0.768$ mm

On substituting the values

$A_2 = 276.96\ mm^2$

$= 461.6\ mm^2$

Now taking smaller value of $A_2 = 276.96\ mm^2$

A_{41} = outward nozzle weld = $(leg)^2 * fr2$

A_{43} = inward nozzle weld = $(leg)^2 * fr2$

From data sheet of nozzle No3, for Weld W302 leg = 9 mm

$A_{43} = A_{41} = 9*9 = 81\ mm^2$

$A_1 + A_2 + A_{41} + A_{43} = 2064.56\ mm^2$

Since $A_1 + A_2 + A_{41} + A_{43} >= A$

Therefore, opening is adequately reinforced.

With Reinforcing Element

A = same as A above = $812.8\ mm^2$

A_1 = same as A1 above= $1625.6\ mm^2$

$A_2 = 5(t_n - t_{rn}) * f_{r2} t$

$= 2(t_n - t_{rn})\ fr2(t_n x2.5 + t_e)$

On Substituting Values

$A_2 = 276.96 mm^2$ (Taking this smaller value)

$= 1625.6\ mm^2$

$A_{41} = leg^2 * f_{r3} = 9*9 = 81\ mm^2$

$A_{43} = leg^2 * f_{r2} = 9*9 = 81\ mm^2$

$A_{41} = leg^2 * f_{r4} = 9*9 = 81\ mm^2$

$A_1 + A_2 + A_{41} + A_{42} + A_{43} = 2145.56\ mm^2$

Since,

$A_1 + A_2 + A_{41} + A_{43} + A_{42} >= A$

Therefore, the reinforcement in the opening is adequate.

NOZZLE

No1

d = 2 inch = 50.8 mm

t_r = 1.99 mm

F = 1

t_n = 10 mm

S_n/S_r = 1.0

F_{r1} = 1

E = 1

$F_{r2} = S_n/S_r$ = 1

F_{r3} = 1

F_{r4} = 1

$A = dt_rF + 2t_nt_rF(1\text{-}Fr1) = 101.6\ mm^2$

1. A1 = 203.2 mm^2

   = 64 mm^2

No2

D = 14" = 355.6 mm

t_r = 1.99 = 2 mm

F = 1

F_{r1} = 1

S_n/S_r = 1.0

t_n = 10 mm

E_1 = 1

$F_{r2} = S_n/S_r$ = 1

F_{r3} = 1

F_{r4} = 1

{t_{rn} = PR/ (Se - 0.6P) = 90*7/ (23879-06*90) = 0.67 mm^2

R = ½*(Finish diameter of circular opening) = 17.88 mm}

$A = dt_rF + 2t_ntF_r(1\text{-}F_{r1})$

$A = 711.2\ mm^2$

$A_1 = d(E_1t\text{-}Ftr) - 2t_n\ (E_1t - Ft_r(1\text{-}F_{r1})) = 1422.4\ mm^2$

$A_2 = 279.69\ mm^2$ (taking this smaller value)

$= 466.15\ mm^2$

A_{41} = Outward nozzle weld = $(leg)^2{*}F_{r2}$

$A_{41} = A_{43} = 81\ mm^2$

$A_1 + A_2 + A_{41} + A_{43} = 1864.09\ mm^2$

Since $A_1 + A_2 + A_{41} + A_{43} + A_{43} >= A$

Therefore, the reinforcement in the opening is adequate.

Vertical Vessels with Skirt

F_g – Total vessel empty dead weight = 17.36KN

P – Design Pressure = 0.62MPa (90Psi)

ΔF_g - Head weight below section 2-2 = 1.572KN

F_f = Weight of vessel content (Liquid) = 4.951KN

$F_g + F_f$ (Vessel Gross weight) = 22.311KN

D_z - Skirt mean Diameter = 1.059m

E_z – Skirt Analysis thickness (Corroded) = 6mm

H – Skirt Height = 1.051m

F_1 – Force at section 1-1 = 20.739KN

M_1 – Moment at section 1-1 = 5154.94Nm

M_b – Applied Moment at bottom of the skirt = 11055.8Nm

W – Weight above skirt = 19.755KN

D_b = Main Shell Diameter = 1.06m

Checking forces and moments at connection areas (Section 1-1, 2-2, 3-3, 4-4)

F_z – Section for different positions wherein subscript ‘p’ indicates moment weakens the load component and subscript q indicates moment strengthens the load component

$F_{zp} = F_1 – F_g - F_f + 4M_1/D_z$

$= -23.579KN$

$F_{zq} = -F_1 – F_g – F_f – 4M_1/D_z = -62.503KN$

a) Membrane Stresses (Here σ represents membrane stresses)

$\sigma_{1p} = (F_{zp} + F_g + F_f)/ (\pi*D_b*E_b) + (P*D_b/4*E_b) = 24.99MPa$

$\sigma_{1q} = (F_{zq} + F_g + F_f)/ (\pi*D_b*E_b) + (P*D_b/4*E_b) = 24.519Mpa$

b) Bending Stresses (Here σ represents Bending stresses)

$M_p = a*F_{zp}$

a = eccentricity of shell wall

$a = 0.5*\sqrt{[(E_b)^2 + (E_z)^2 + 2*E_b*E_z*\cos Y]}$

where $\cos Y = 1- (D_b + E_b – D_z + E_z)/2(r+E_b) = 0.9648$

a = 5.9645mm

$\sigma_{1q(a)} = (C*6*M_q)/ (\pi*D_b*E_b^2) = -10.659KN$

$\sigma_{3p(a)} = (C*6*M_p)/ (\pi*D_z*E_z^2) = -4.028KN$

$\sigma_{3q(a)} = (C*6*M_q)/ (\pi*D_z*Ez^2) = -10.669KN$

Bending effect due to internal Pressure in the knuckle

$\sigma_{1(p)} = \sigma_{2(p)} = (P_1+P_h) *D_b*((V\alpha /Va)-1))/4*E_b$

where,

$y = 125*E_b/D_b = 0.7075$

α = 4.2 – 0.2y = 4.058 (For Elliptical Ends)

σ = 158.68Mpa

Total Stress at Different Sections

Section 1-1

$\sigma_{1(pi)}$ = 187.694MPa

$\sigma_{1(pi)}$ = -129.66MPa

$\sigma_{1(qi)}$ = 193.85MPa

$\sigma_{1(qo)}$ = -123.502MPa

Section 2-2

$\sigma_{2\,(pi)}$ = 182.365MPa

$\sigma_{2\,(po)}$ = -126.947MPa

$\sigma_{2\,(qi)}$ = 175.73MPa

$\sigma_{2\,(qo)}$ = -120.312MPa

Section 3-3

$\sigma_{3(pi)}$ = 7.535MPa

$\sigma_{3(po)}$ = -13.8083MPa

$\sigma_{3(qi)}$ = 2.845MPa

$\sigma_{3(qo)}$ = -5.211MPa

Chapter 4
Implementation and experimentation of the issues related to project work/topic

This chapter presents the procedure for the further optimizing of the design by various iterations on FEA software and excel sheet.

An excel sheet was made for directly computing various results using ASME codes taking inputs as given in the drawing sheet based on customer needs. Various inputs taken for computing results were internal pressure, external pressure, shell diameter and length of the pressure vessel. Firstly, thicknesses were calculated on the basis of both internal and external pressures separately and thickness was adjusted based on the industry requirements. Then the thickness is checked by considering for pressure on concave and convex side of formed heads. Summarizing, we directly computed thickness of the vessel by providing just the input conditions using Microsoft Excel.

Given:-	Pint. (psi)	90						
	Pext. (psi)	14.5						
	Tmax(F)	400						
	Corrosion alowance	0						
	Main shell material	SA - 240						
	Nozzle material	SA - 240						
	Shell dia.(ID)	41.5						
	Length, L	75.2						
Calculation of t for shell end under external pressure by UG 27					**Calulation of t for shell under external pressure by UG 28**			
For circumferential stress			***For longitudinal stress***		Do	41.97244		
S(max. allowable stress) psi.		23879			L/D0	1.791651	Do/t	177.6833
(by ASME BPVC II D pg.126)					From ASME BPVC II D.M Subpart 3 factors A & B are determined			
E(joint eff. for full type)		1			From Pg. 794			
t = (pR)/(SE-0.6p)		0.078384	t = (pR)/(2SE+0.4P)	0.03907394	A	0.00032		
	in mm.	1.990954		0.992478135	From Pg. 797(for σy > 207MPa			
					B	33		
Considering Industry requirements					Pa	max. allowable external working pressure		
t (in mm.)	6				Pa = (4B)/(3*Do/t)		0.24763155	
t (in inch.)	0.2362204724				Pa(in psi.)		35.9159204	
					As Pa>Pexternal		Safe	

Fig. 3 Excel sheet for shell and dished end thickness

The finite element analysis was performed on ANSYS workbench. The solid model was developed on Solidworks.

Iteration1:

The first iteration will check the structural integrity of the pressure vessel under internal and external pressure as specified by the company's client. The thickness used is the one which was arrived at using analytical calculations using ASME codes. This iteration involves the modelling of only the shell and dished end. Further iterations will aim to analyse the effects of adding supports and nozzles.

Fig. 4 The solid model consisting the shell and dished ends

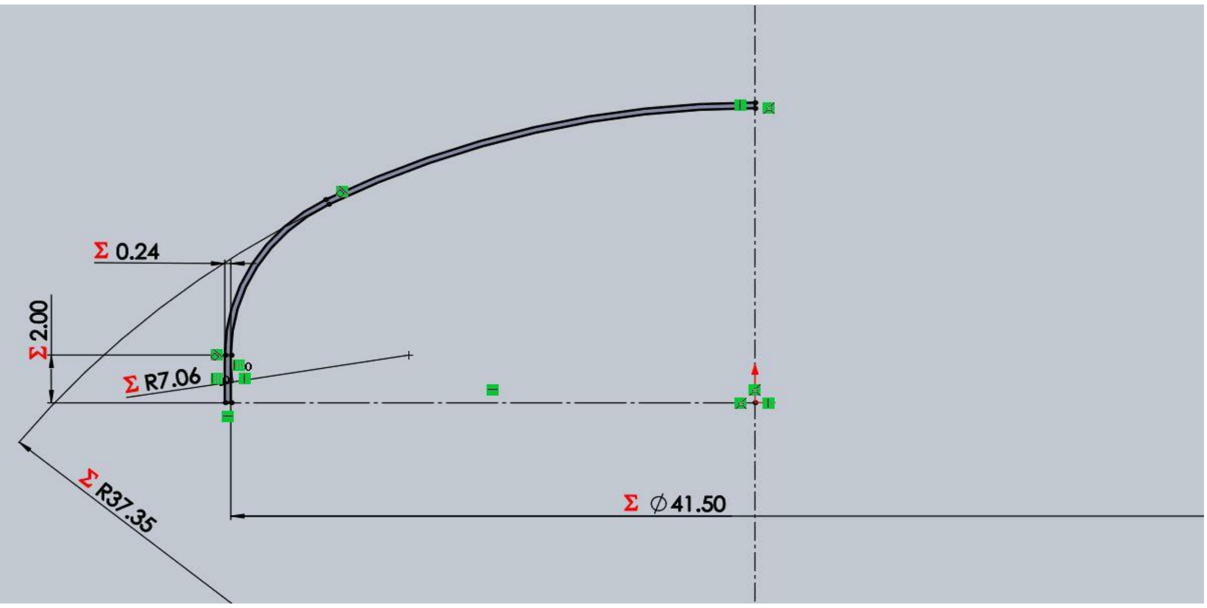

Fig. 5 2-D drafting on Solidworks

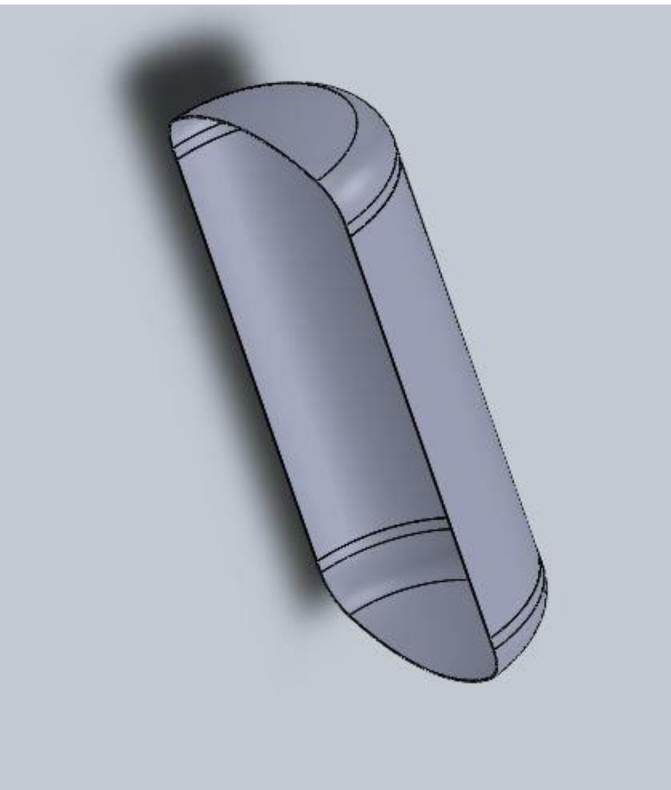

Fig. 6 The solid model (sectional view)

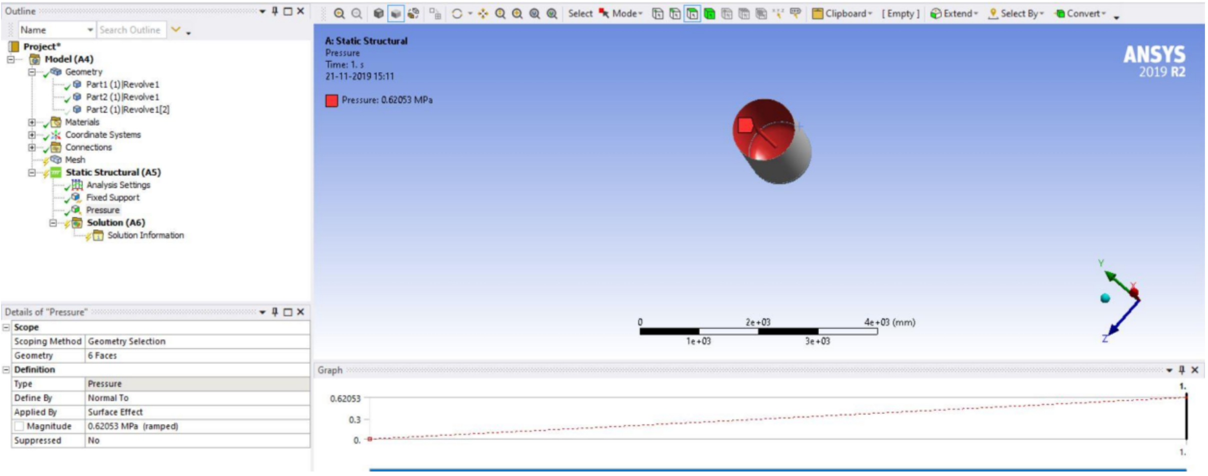

Fig. 7 The boundary conditions

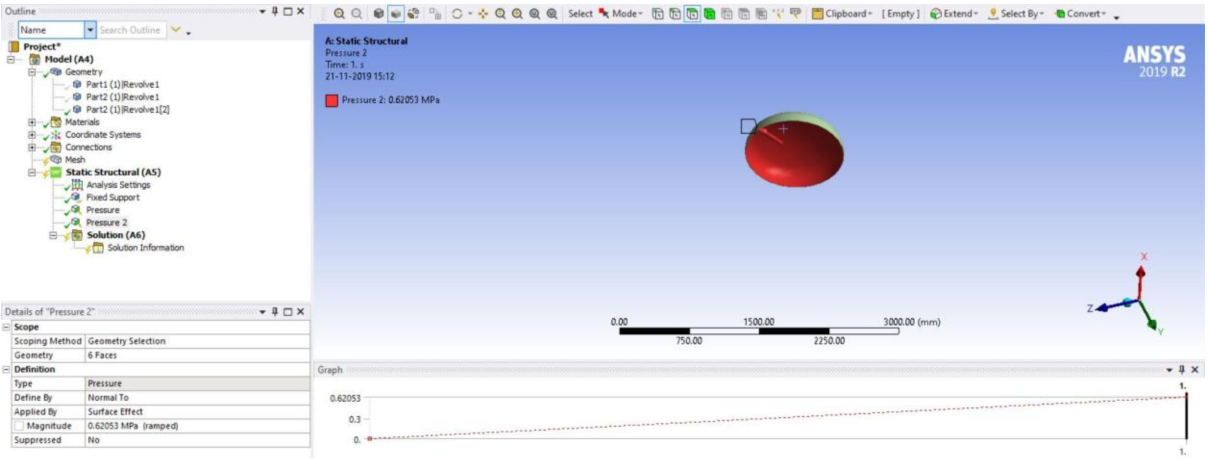

Fig. 8 The boundary conditions (fixed support)

The internal pressure of 0.62 MPa was applied (in red) in the outward direction as per the customer requirements.

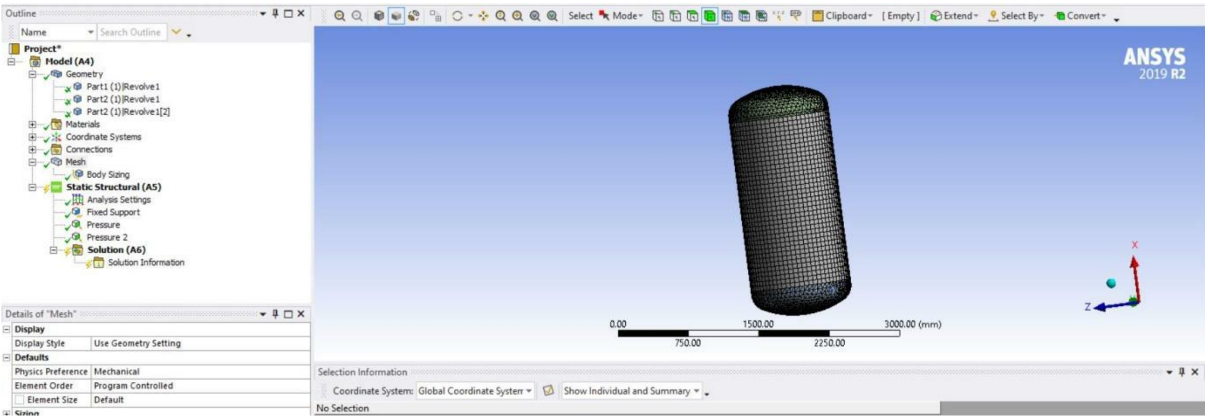

Fig. 9 Meshing

During the meshing process, 21511 Nodes and 10465 Elements were obtained according to the statistics.

Iteration I: FEA Analysis of Pressure Vessel without Nozzles and Skirt Support.

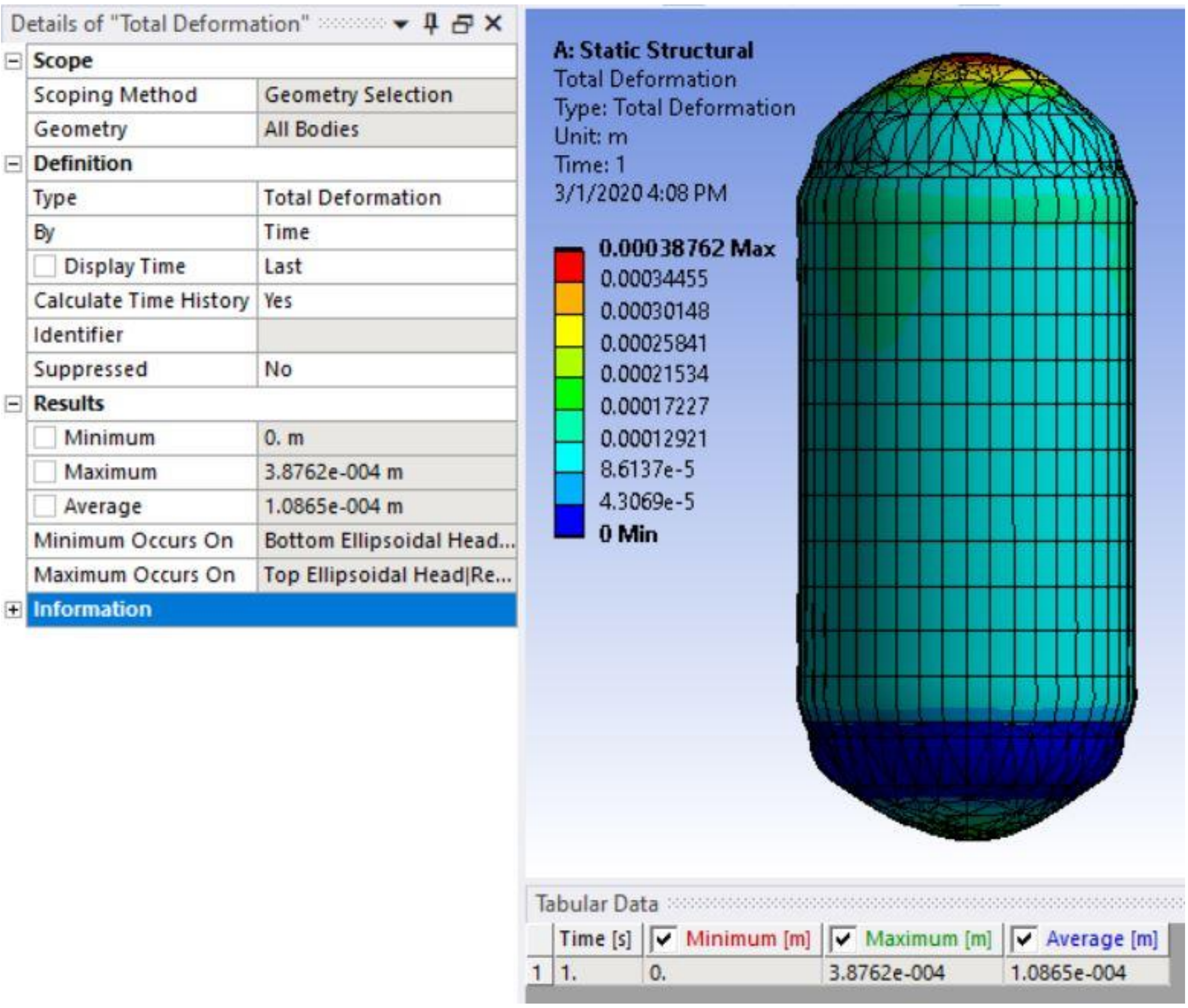

Fig. 10 Total Deformation for 6mm pressure vessel

The maximum deformation of 0.388mm was obtained for the 6mm pressure vessel in the first iteration (without nozzles and skirt support).

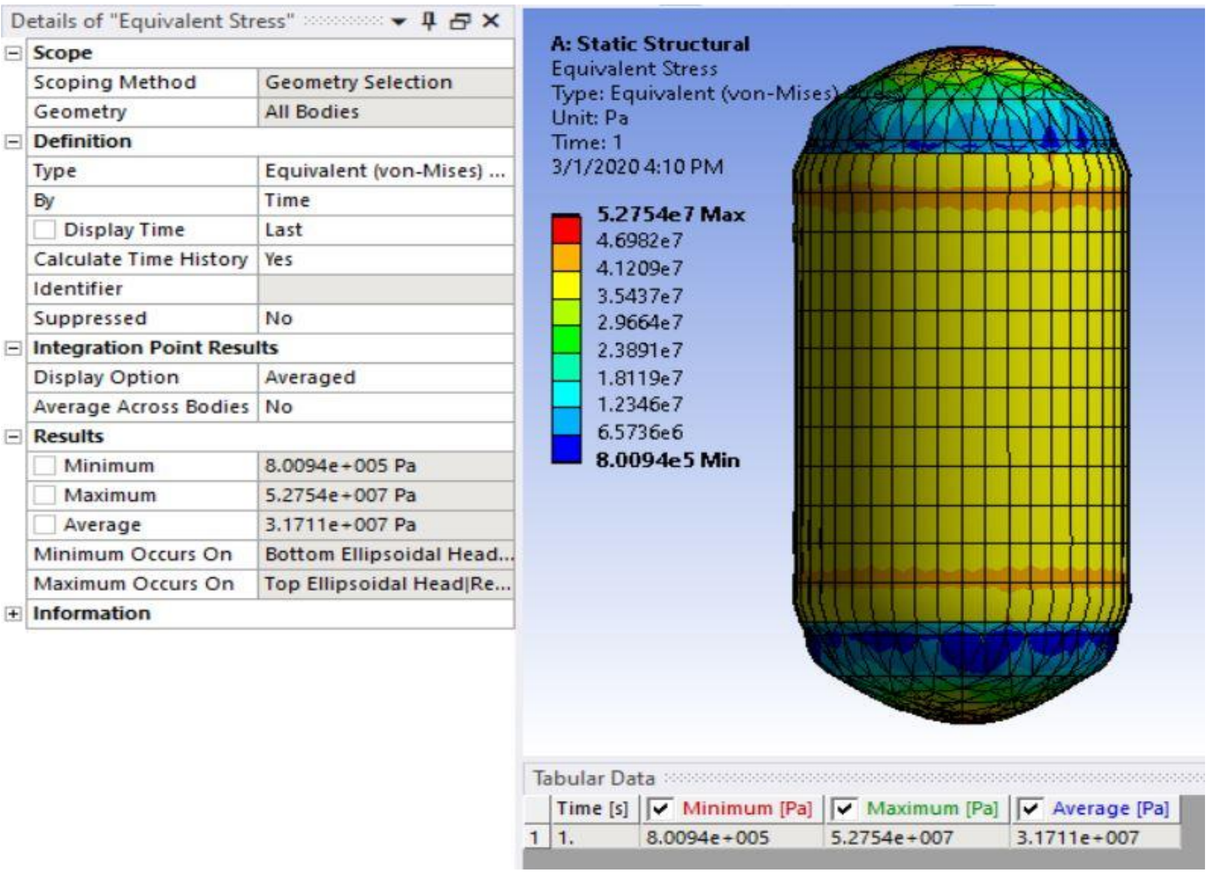

Fig. 11 Equivalent stress for 6mm pressure vessel

The maximum equivalent (von-Mises) stress of 52.75 MPa was obtained for the 6mm pressure vessel in the first iteration (without nozzles and skirt support).

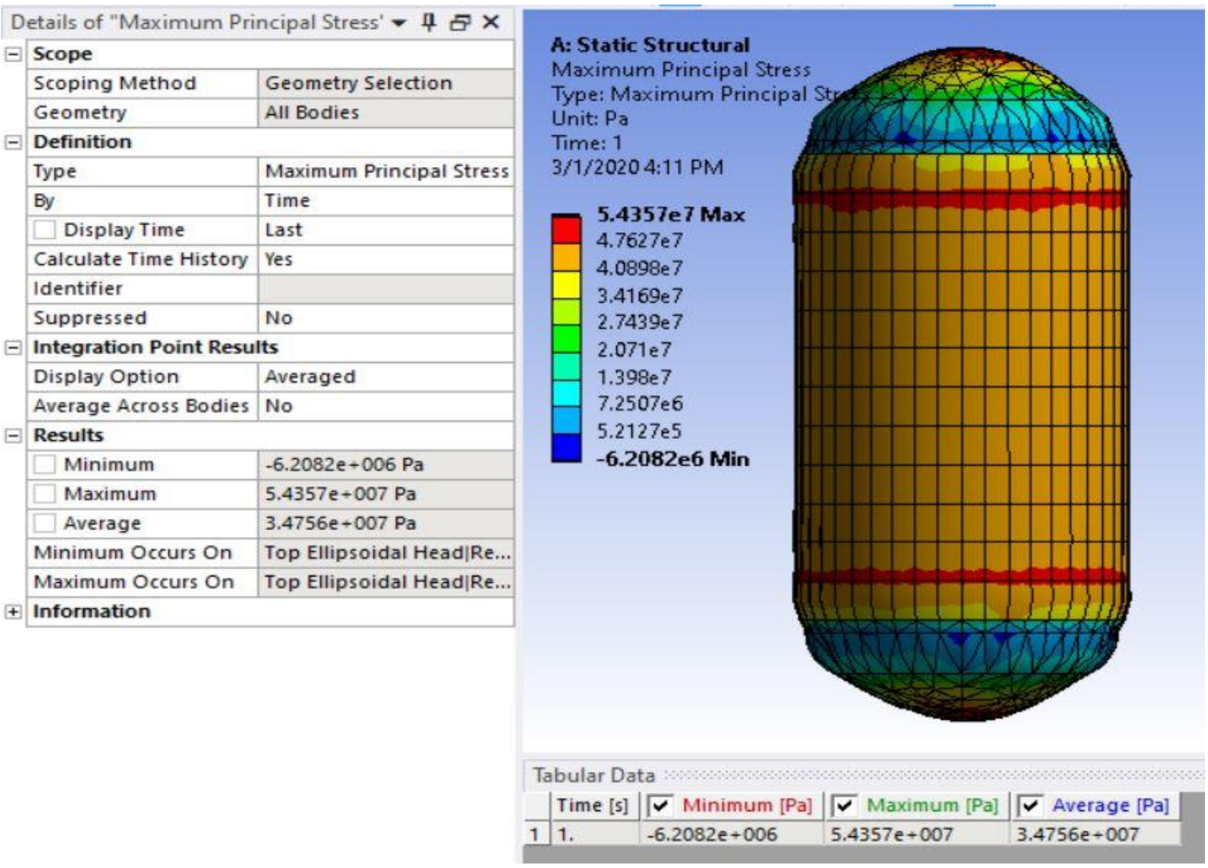

Fig. 12 Maximum principal stress for 6mm pressure vessel

The maximum principle stress of 54.35 MPa was obtained for the 6mm pressure vessel in the first iteration (without nozzles and skirt support).

Table 2 Iteration I FEA Results for various thickness

Sr No	Description	Maximum Equivalent Stress (MPa)	Maximum Deformation (mm)
1	2 mm thickness	253.1	1.643
2	3 mm thickness	161.09	1.011
3	4 mm thickness	108.32	0.669
4	5 mm thickness	73.95	0.489
5	6 mm thickness	52.75	0.387

Iteration II: FEA Analysis of Pressure Vessel with Top Nozzles only.

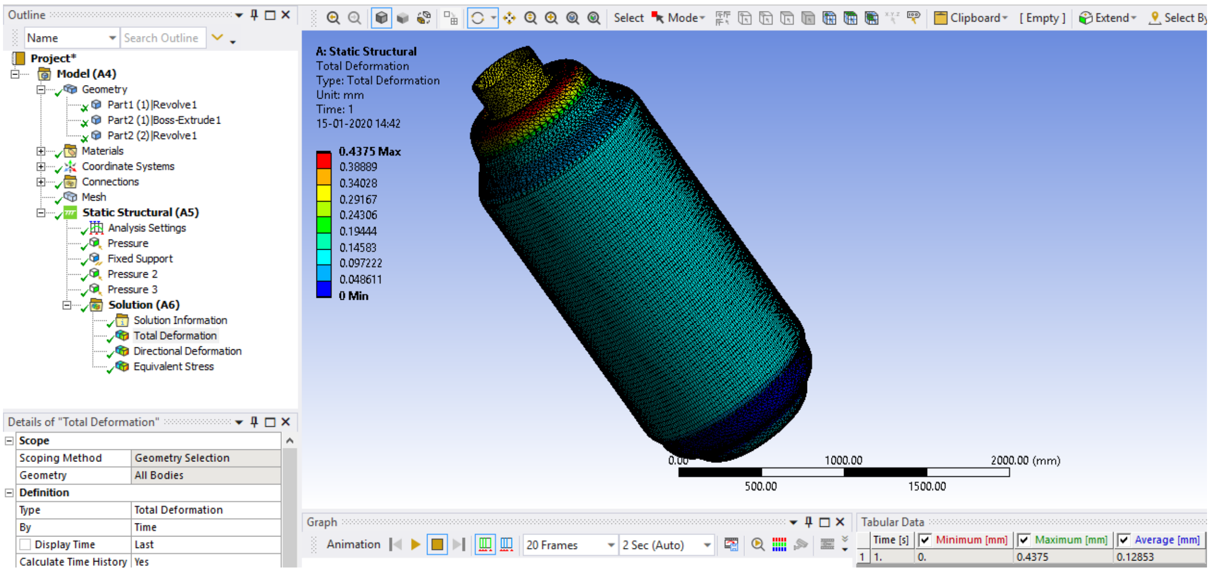

Fig. 13 Total Deformation with single nozzle only for 6mm

The maximum deformation of 0.44 mm was obtained for the 6mm pressure vessel in the second iteration (with top nozzle only). As seen from the above result, the region of maximum stress concentration (in red) is at the junction of the attachment of the top nozzle to the dished end of the pressure vessel.

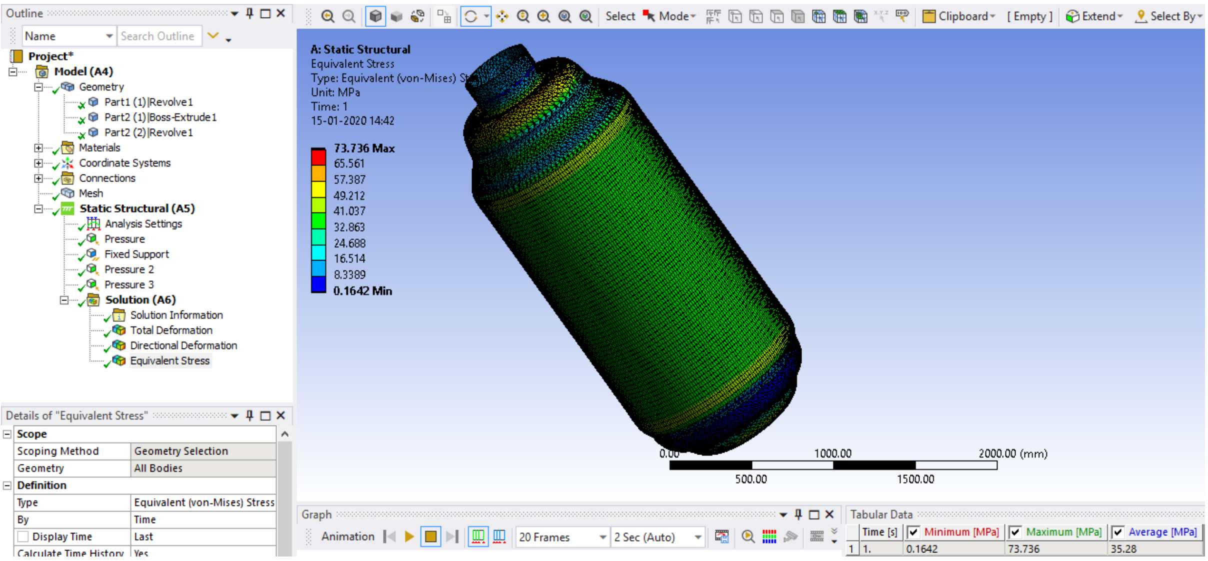

Fig. 14 Equivalent (von-Mises) stress with single nozzle only for 6mm

The maximum Equivalent (von-Mises) stress of 73.74 MPa was obtained for the 6mm pressure vessel in the second iteration (with nozzles and skirt support).

Iteration III: FEA Analysis of Pressure Vessel with Skirt Support only.

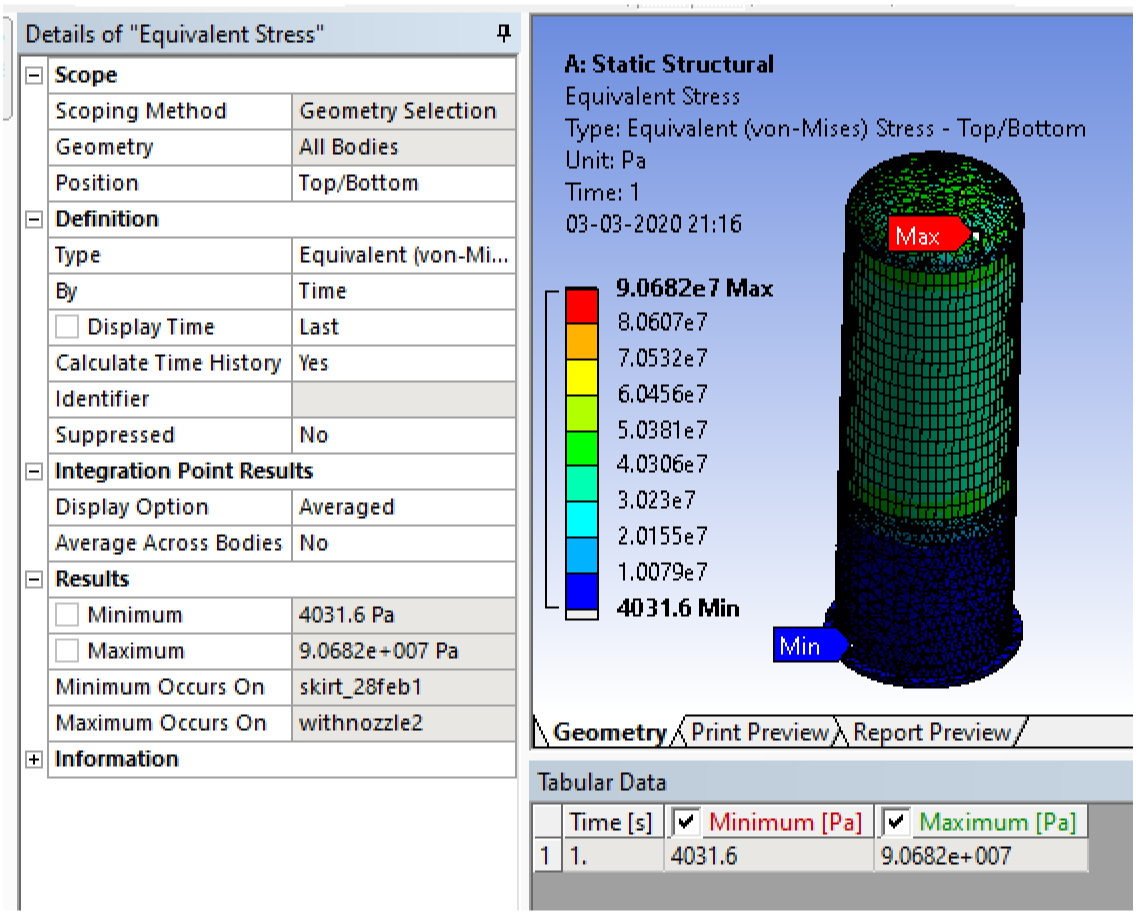

Fig. 15 Equivalent stress with skirt support only for 6mm

The maximum equivalent (von-Mises) stress of 90.07 MPa was obtained for the 6mm pressure vessel in the third iteration (with skirt support only).

Iteration IV: FEA Analysis of Pressure Vessel with Nozzles and Skirt Support.

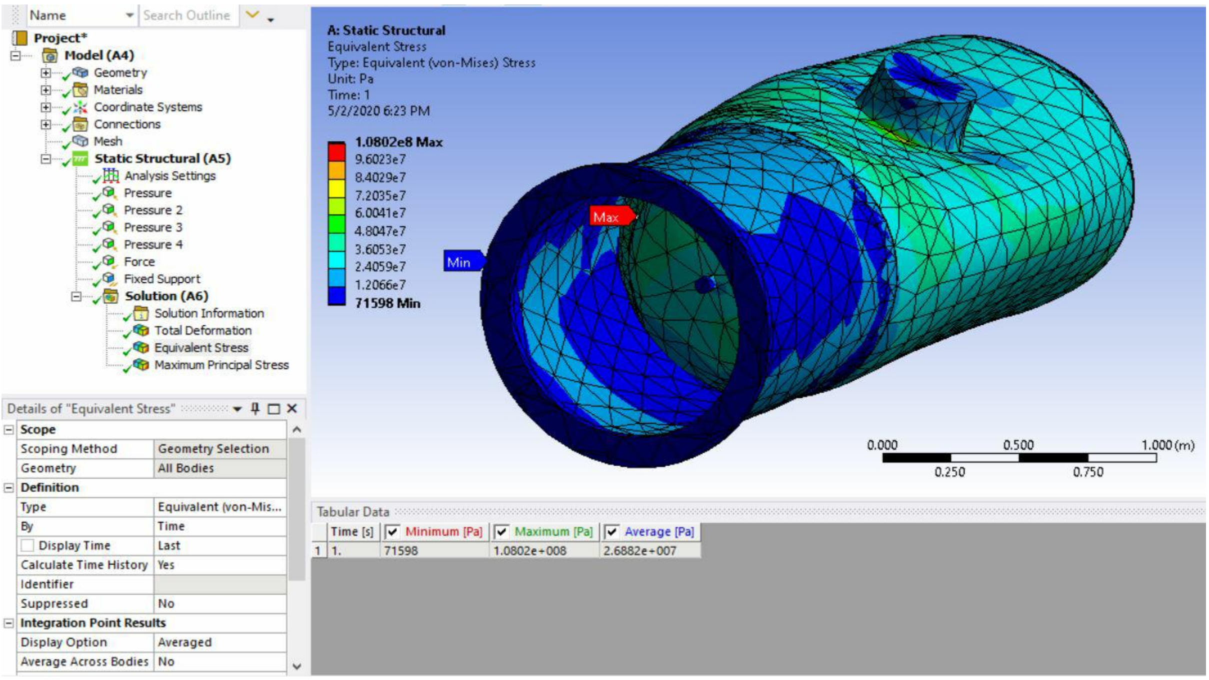

Fig. 16 Equivalent stress with maximum and minimum probe points

The maximum equivalent (von-Mises) stress of 108.02 MPa was obtained for the 6mm pressure vessel in the second iteration (with nozzles and skirt support).

As expected, the value of stress in the second iteration, with addition of nozzles and skirt support has almost doubled from 52.75 MPa to 108.02 MPa due to the addition of sharp and sudden edges between the nozzle, skirt support and shell joints, making it areas for high stress concentration and accumulation.

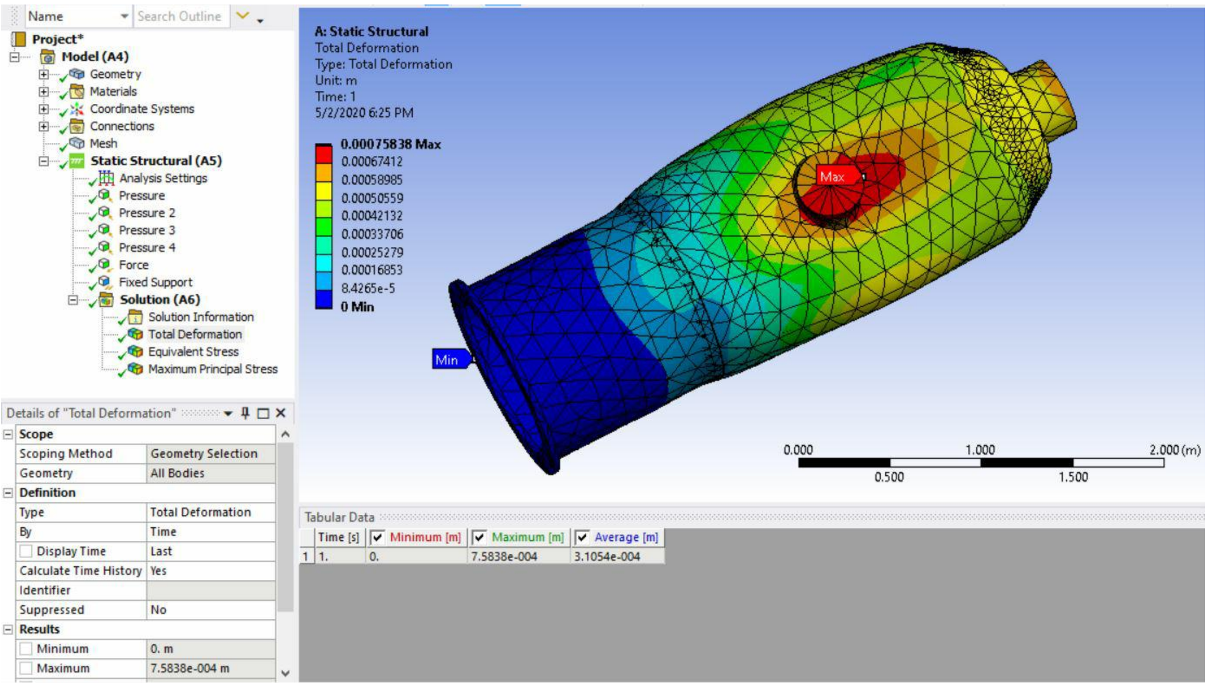

Fig. 17 Total deformation with maximum and minimum probe points

The maximum total deformation of 0.76 mm was obtained for the 6mm pressure vessel in the second iteration (with nozzles and skirt support).

In this case too, the deformation has increased two folds but is within the acceptable limits as per the requirements and hence the design is safe.

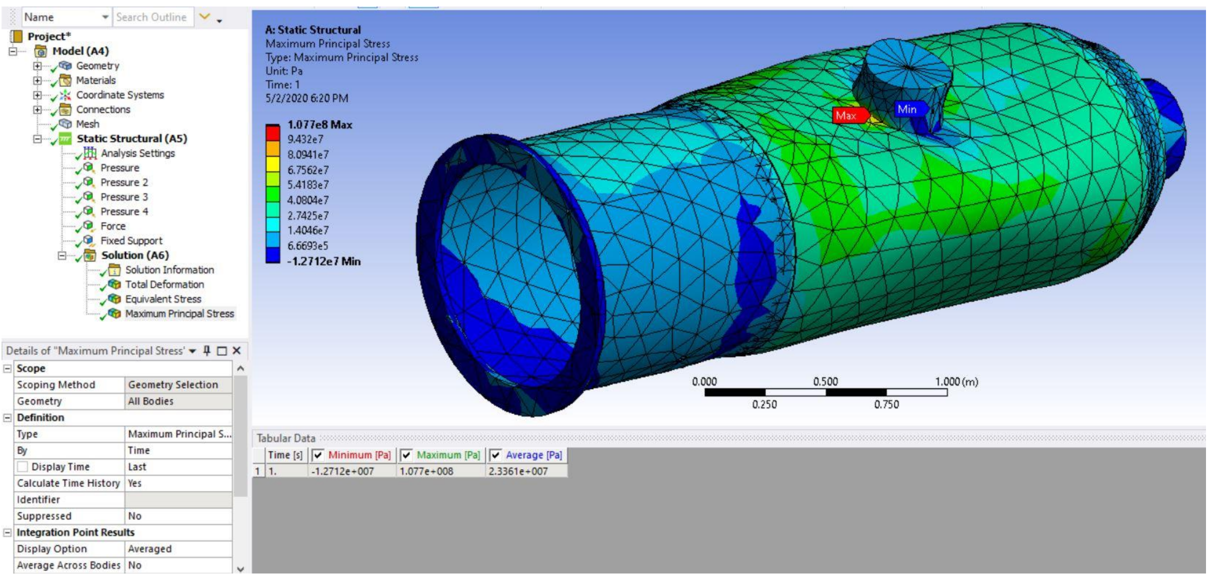

Fig. 18 Maximum principal stress with maximum and minimum probe points

The maximum principle stress of 107.7 MPa was obtained for the 6mm pressure vessel in the second iteration (with nozzles and skirt support).

A similar trend is seen where the stress values increase by two times as compared to the first iteration but is below the allowable stress value for the used material at the operated temperature of 204.4 C

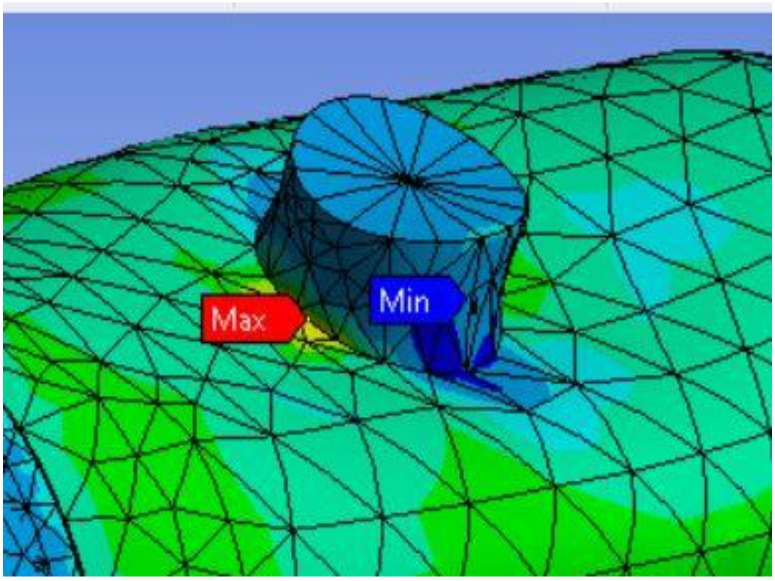

Fig. 19 Location of maximum principal stress with maximum and minimum probe points

Table 3 Final summary of the FEA results for all the four iterations

Sr No	Description	Maximum Equivalent Stress (MPa)	Minimum Equivalent Stress (MPa)	Maximum Deformation (mm)
Iteration 1	Without nozzle and skirt	52.754	0.801	0.3876
Iteration 2	With top nozzle	73.76	0.16	0.4375
Iteration 3	With skirt support	90.682	0.0043	0.503
Iteration 4	With nozzle and skirt	107.02	0.0716	0.78

Various Plots based on the final FEA results

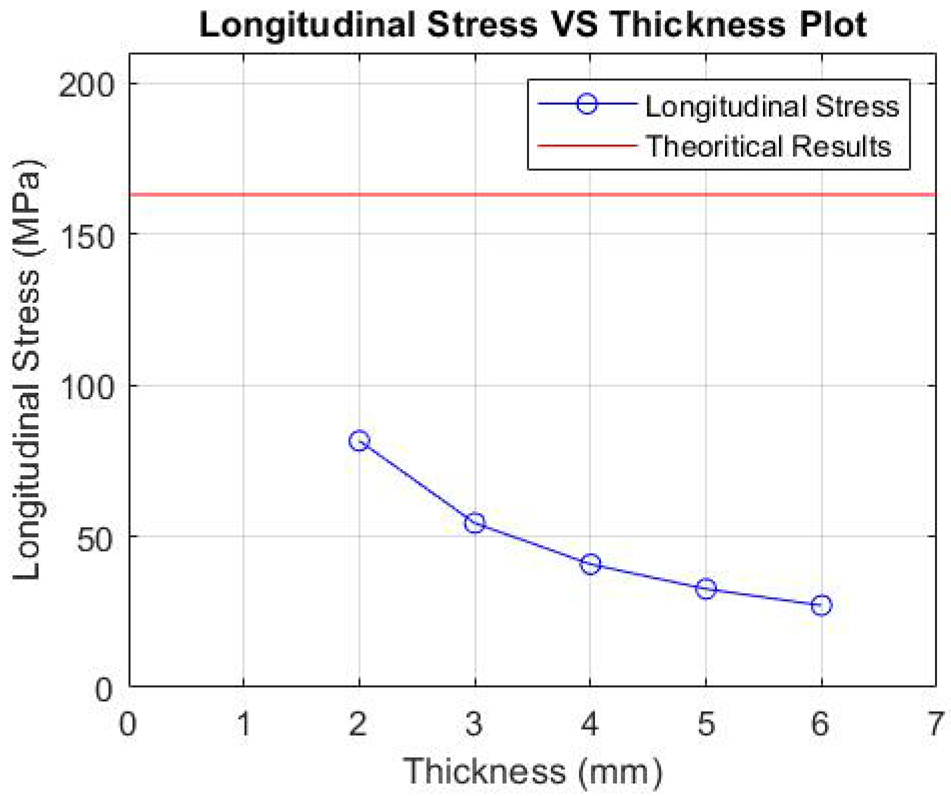

Fig. 20 Plot for longitudinal stress V/S Thickness for 6mm

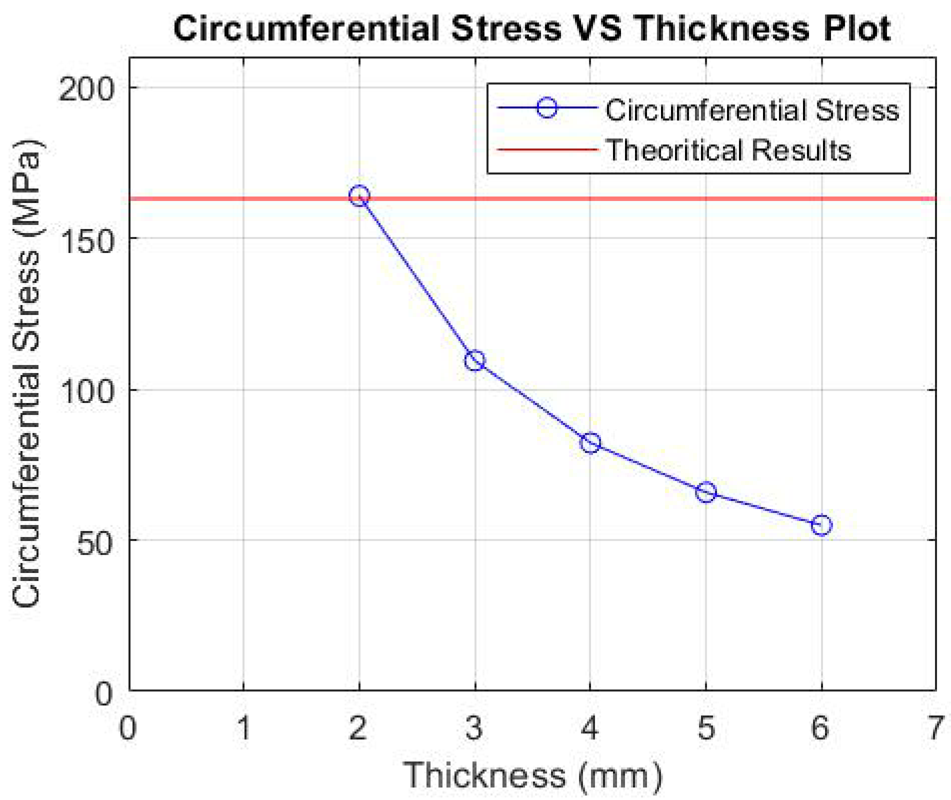

Fig. 21 Plot for circumferential stress V/S Thickness for 6mm

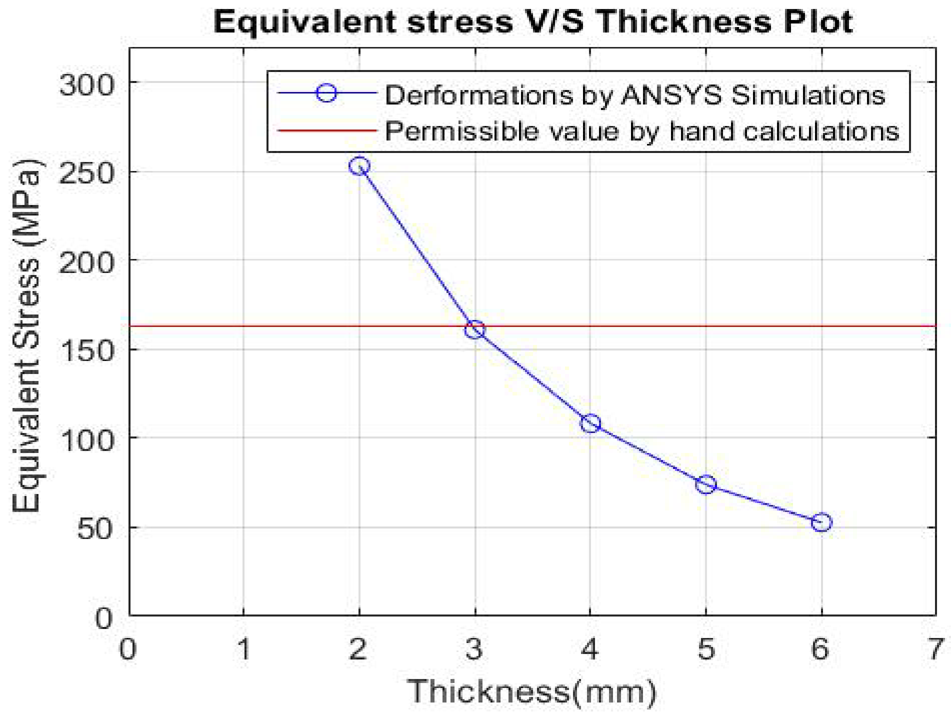

Fig. 22 Plot for Equivalent stress V/S Thickness for 6mm

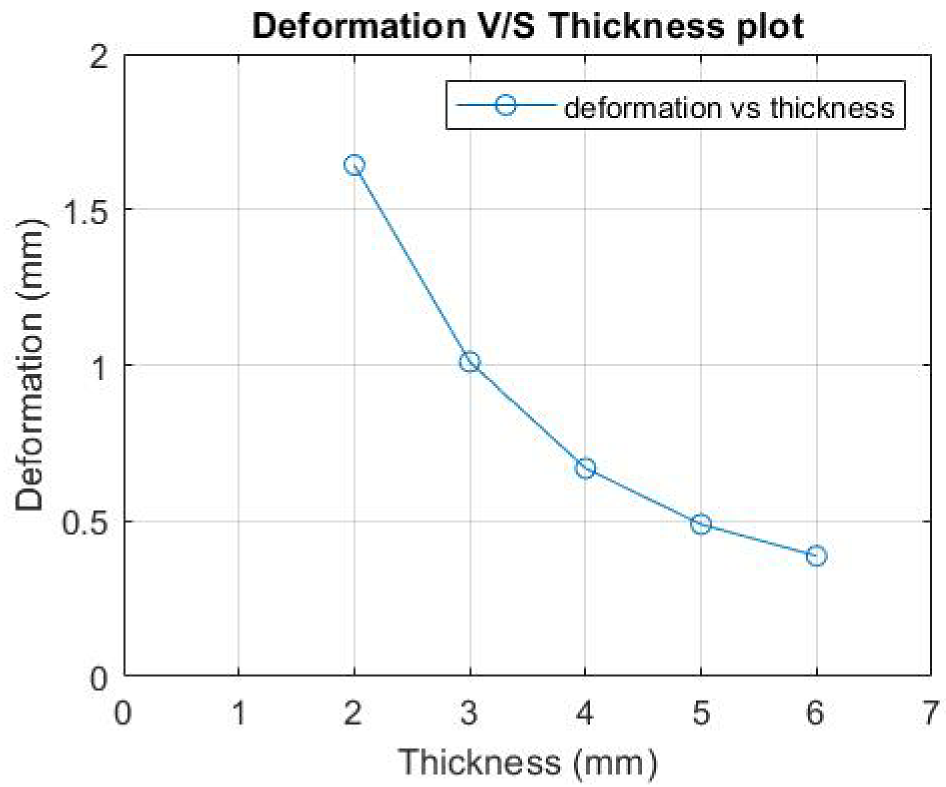

Fig. 23 Plot for Deformation V/S Thickness for 6mm

From the above graphs for various Stress vs Thickness, it is visible that the value of stress decreases exponentially as the thickness increases from 2mm to 6mm. All the obtained values from the FEA results are below the allowable value of Stress (163 MPa), indicating that the design is safe and acceptable.

Chapter 5
Conclusion and further work

This chapter presents the conclusion after analysing the theoretical calculations and FEA results. Further work regarding future iterations is included

5.1 Conclusions:

The design calculations of vertical pressure vessel were done using ASME codes and the analysis of same was done using FEA software. The calculations for following parameters were carried out:

- Thickness of Shell and Dished end
- Longitudinal and membrane stresses in the vessel
- Reinforcing thickness for nozzle openings and their safety Criteria
- Thickness of Skirt Support
- Membrane stresses in various sections of the skirt support

Subsequently, the vessel was modelled in Solidworks and FEA was carried out in Ansys 19. Static structural analysis was done. Analysis of different thickness of the shell was carried out from 2 mm (minimum calculated thickness) to 6mm (thickness approved in the industry). The equivalent stresses and deformation show a rising trend with decrease in thickness. The circumferential and longitudinal stresses obtained by FEA analysis are less than those calculated by ASME codes. This can be explained by the basic modelling of the vessel and coarse meshing due to limitations of computational power.

In subsequent iterations, the openings for Nozzle and the skirt support were modelled and the static structural FEA analysis was done. In these iterations maximum stress concentration is seen near the openings for nozzle indicating that the vessel was weakened in the parts near the openings. The equivalent stresses obtained by FEA were compared with maximum allowable stresses for the shell, dished and skirt material at the operating temperature. The maximum equivalent stresses were found to be below the limits of maximum allowable material stresses, thus indicating safe design. Also, the maximum stress in the Skirt support is less than that in the vessel. This could be explained based on the fact that the skirt material has higher value of elastic modulus.

Based on the static burst pressure analysis alone, it could be concluded that the vessel maintains structural integrity also at 5mm thickness i.e. less than that as suggested during the industrial design procedure-6mm. However, a more detailed model of the vessel including all the openings for nozzles, manholes and its supporting structures like skirt, base ring, gusset plate etc. is needed for detailed analysis. Further, analysis of thermal stress is also necessary at the operating temperature to get a realistic idea of stresses in the vessel, which is beyond the scope of this thesis. Hence the design thickness of 6mm was inculcated in the fabrication by the company.

5.2 Further Work

A few areas where there is scope for further work in this project are:

- Thermal FEA analysis can be done at the operating temperature of the Pressure vessel to see the effects of Thermal Stresses on the vessel
- The current FEA analysis could be refined with using a finer mesh, especially around critical areas such as openings for nozzles, in order to get a better representation of stress concentrations in the vessel.
- A different FEA solver could be utilized with greater computing power to generate fine mesh to see the variation of stresses across the thickness of the vessel.
- A computer program could be developed which, when fed the results of detailed FEA analysis can attempt to optimize the design at lesser thickness and also be able to calculate economic savings by doing so.

Chapter 6
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